RecoMart: Product Recommendation Pipeline

Problem Formulation Report

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1. Business Problem Definition

RecoMart is an e-commerce platform experiencing suboptimal product discovery and low conversion rates. Customers frequently browse without purchasing because they are not presented with products relevant to their preferences and behaviour. The core business challenge is to increase conversion rates, average order value, and customer retention by surfacing the right products to the right users at the right time.

The solution is an automated, end-to-end data management and machine learning pipeline that ingests product and user interaction data, engineers meaningful features, trains recommendation models, and makes model-ready features available for real-time inference.

1.1 Business Objectives

- Improve product conversion rate by surfacing personalized recommendations
- Increase average session depth by reducing irrelevant product exposure
- Enable data-driven merchandising decisions using product quality scores
- Build a scalable, automated pipeline that re-trains on fresh interaction data
- Provide explainable recommendations using content-based feature signals

1.2 Success Criteria

Metric	Target	Measurement Method
Precision@10	>= 0.15	Evaluated on holdout user-item pairs
Recall@10	>= 0.10	Fraction of relevant items retrieved
NDCG@10	>= 0.20	Ranking quality of top-10 recommendations
Content-Based R2	>= 0.85	GBR regression on recommendation probability
Data Quality Score	100%	All automated validation checks pass

2. Data Sources and Attributes

2.1 Source 1 - Kaggle Ecommerce Product Dataset

Dataset: "Ecommerce Product Recommendation Dataset" by Kartikey Bartwal (Kaggle)

Ingestion: kagglehub.dataset_download() with retry logic | 1,018 product records after deduplication

Column	Type	Description
Number of clicks on similar products	Float	Click engagement signal
Number of similar products purchased	Float	Purchase conversion signal
Average rating given to similar products	Float	Social proof signal
Gender	Categorical	User gender (label-encoded)
Median purchasing price (in rupees)	Float	Price sensitivity signal

Rating of the product	Float	Direct product quality score
Brand of the product	Categorical	Brand identity (label-encoded)
Customer review sentiment score	Float	NLP-derived sentiment [-1, 1]
Price of the product	Float	Actual product price (INR)
Holiday / Season	Categorical	Temporal demand context
Geographical locations	Categorical	Regional preference signal
Probability for recommendation	Float	TARGET: recommendation label

2.2 Source 2 - Simulated REST API (User Interactions)

Synthetic dataset simulating real-time API-delivered user behaviour logs.

5,000 interactions | 200 users | Last 90 days of timestamps

Column	Type	Description
user_id	String	User identifier (U001-U200)
product_id	Integer	Product reference from Kaggle dataset
action_type	Categorical	view / click / purchase / rate
rating	Float	Explicit rating 1.0-5.0 (for rated interactions)
timestamp	Datetime	Interaction time (last 90 days)

3. Pipeline Architecture and Expected Outputs

The pipeline is orchestrated end-to-end by Prefect, with 7 sequential tasks. Each task produces clearly defined outputs consumed by the next stage.

Pipeline Stage	Key Output
Stage 1: Data Ingestion	Raw CSV files partitioned by source/type/date in data/raw/
Stage 2: Data Validation	HTML + PDF quality report; validated CSVs with 100% pass rate
Stage 3: Data Preparation	Cleaned CSVs; label encoders; MinMaxScaler; 6 EDA plots
Stage 4: Feature Engineering	Product + User feature CSVs, Parquet files, SQLite warehouse
Stage 5: Feature Store (Feast)	Registered feature views; materialized SQLite online store
Stage 6: Data Versioning	SHA-256 version manifest; data lineage graph (JSON)
Stage 7: Model Training	Trained GBR + SVD models; MLflow experiment runs logged

3.1 Recommendation Models

Model	Algorithm	Input	Output
Content-Based	GradientBoostingRegressor	Product features (15)	Recommendation probability
Collaborative	SVD (Surprise lib)	User-Item ratings matrix	Predicted rating

4. Evaluation Metrics

4.1 Ranking Metrics (Collaborative Filtering)

Metric	Definition	Significance

Precision@K	Fraction of top-K recommended items that are relevant	Measures recommendation accuracy
Recall@K	Fraction of all relevant items appearing in top-K	Measures recommendation coverage
NDCG@K	Discounted Cumulative Gain - rewards highly ranked relevant items	Measures ranking quality

4.2 Regression Metrics (Content-Based)

Metric	Description	Value	Direction
RMSE	Root Mean Squared Error	0.0769 (achieved)	Lower is better
MAE	Mean Absolute Error	0.0454 (achieved)	Lower is better
R2	Coefficient of Determination	0.9361 (achieved)	Closer to 1.0 is better

5. Technology Stack

Component	Technology
Language	Python 3.x
Orchestration	Prefect (DAG with 7 tasks, retry logic)
Feature Store	Feast (SQLite offline + online store)
Experiment Tracking	MLflow (SQLite backend)
Data Warehouse	SQLite (via pandas + SQLAlchemy)
Content-Based Model	Scikit-learn GradientBoostingRegressor
Collaborative Model	Surprise SVD
Validation	Custom pandas-based profiler
Visualization	Matplotlib + Seaborn
Version Control	Git + GitHub